Workshop Pole Barn: Plans and Elevations

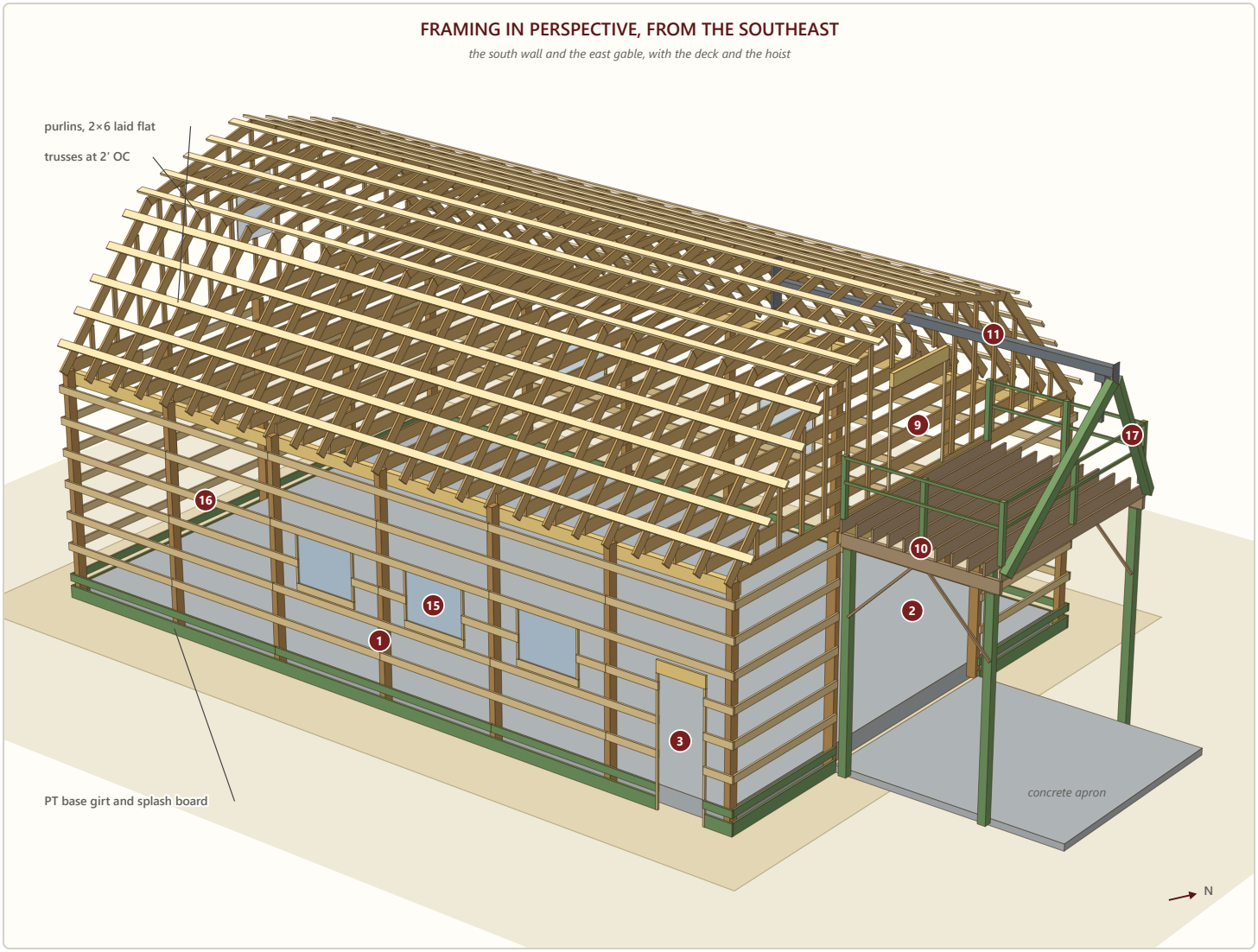
Floor plan, loft plan, the four elevations, the framing in perspective, and the wall framing · 32 × 48 ft · 14 ft eave · gambrel · numbered markers match the legend below

Drawn from the design model at 32 × 48 ft, so the posts, trusses, doors, and windows fall where the dimensions say. The drawings show the design intent for the engineer and the truss plant; they are not construction documents. North is up in the plans. The compass and the scale bar give the direction and the size.

Build Marker Legend

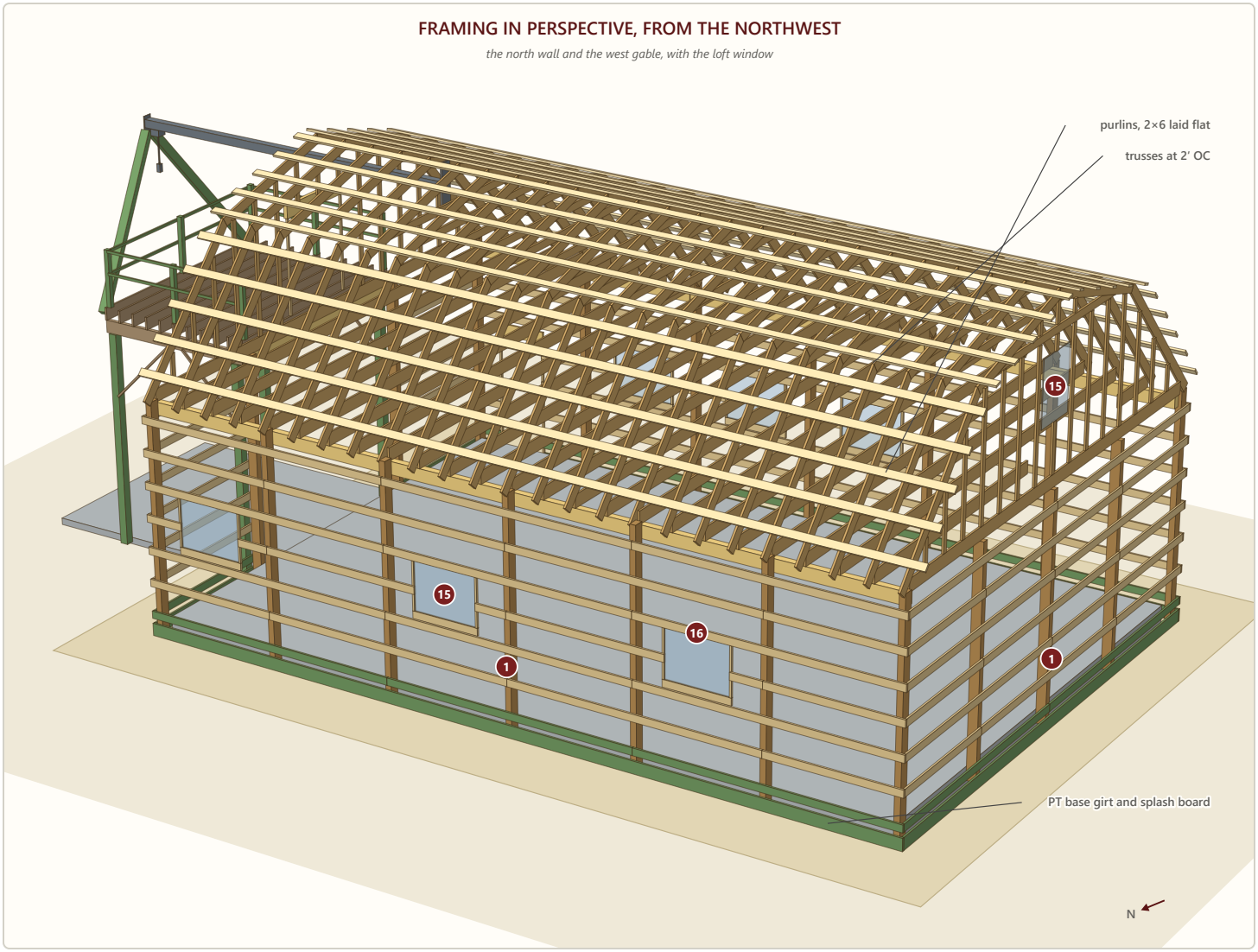
Marker	Component	Notes
1	3-ply 2×8 laminated post on a wet-set bracket	Sidewalls at 8-8-8-8-8-8 ft; the west gable at 8 ft; two door-jamb posts on the east gable
2	14′ × 12′ insulated overhead door	Wind rated for 115 mph, R-12 or better, low-headroom track, centered on the east gable
3	3070 insulated walk door	South sidewall, near the east end
4	Stair landing, 4′ × 4′, at the southwest corner	Between the first and second flights, 6′-5″ above the floor
5	Loft floor opening, about 4′ × 11′, at the west wall	Over the whole second flight, where it enters the loft: tell the truss plant
6	Wood stove pad (location to be chosen)	Non-combustible base; Class A chimney through the wall or the roof
7	200 A electrical sub-panel	East wall inside, south of the main door
8	Mini-split condenser pad	Outside the west or the south wall (the owner's pick)
9	5′ × 6′ loft double doors	East gable, centered above the main door at loft level
10	14′ × 10′ landing deck at loft level	Free-standing on 4 PT posts to pier footings
11	Hoist beam (W6 I-beam) through the east gable	Reaches 12′-6″ east of the wall, 2 ft past the deck's outer edge, and runs 9 ft inside; 1-ton chain fall, 25 ft lift
12	Concrete apron (camper approach)	14 ft wide, 10–20 ft east of the main door
13	200 A sub-feed conduit	Underground, from the south (the pump house is about 29 ft away); the source is to be confirmed
14	Snow guards on the lower slope	Over the walk door, on the south slope
15	Windows: 7 in all	3 in the north wall and 3 in the south wall, 4′ × 3′ sliders with the sill 5′ above the floor; one 3′ × 4′ single-hung in the west gable at the loft
16	2×6 SPF #2 girts, laid flat on the outside face of the posts	24″ OC: 7 rows on each wall, the lowest a pressure-treated base girt; 16 ft boards with every joint over a post (north and south walls, east and west walls)
17	Hoist braces: two raking 6×6 posts	From the deck's outer corners up to the tip of the hoist beam, about 11′-9″ each; the hook hangs between them (in perspective)

The Framing in Perspective



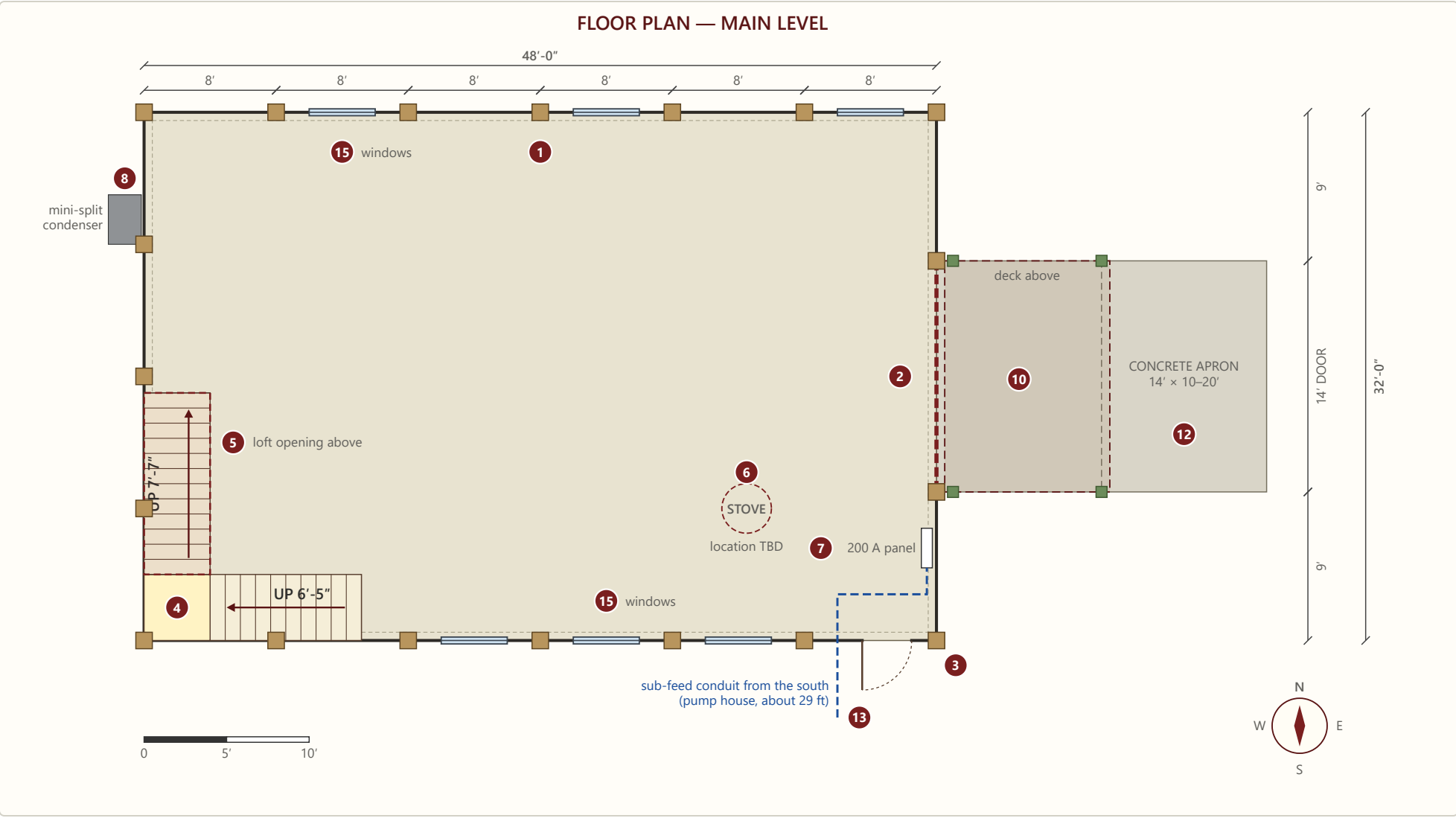
The frame from the southeast, with the steel, furring, sheathing, and decking left off so that every member shows. The **posts** (marker 1) stand 8-8-8-8-8-8 ft apart on the sidewalls, and the **2×6 girts** (marker 16) lie flat across their outside faces at 24″ OC, with the treated base girt and the 2×10 splash board at the foot. The LVL truss carriers ride on the inside of the sidewall posts and carry the 25 gambrel **trusses** at 2′ OC, whose bottom chords are the loft joists. A **collar tie** joins the two upper chords of each of T1 to T20, 8′-0″ above the loft floor; the five hoist trusses at the east end have none. The 2×6 **purlins** lie flat on the top chords, five rows on each slope. At the east end the deck (marker 10) stands free of the wall on four posts, and the hoist beam (marker 11) runs out through the gable past the deck, held at its tip by the two braces (marker 17). The truss webs are not drawn: the truss plant designs them.

The Framing in Perspective: Northwest



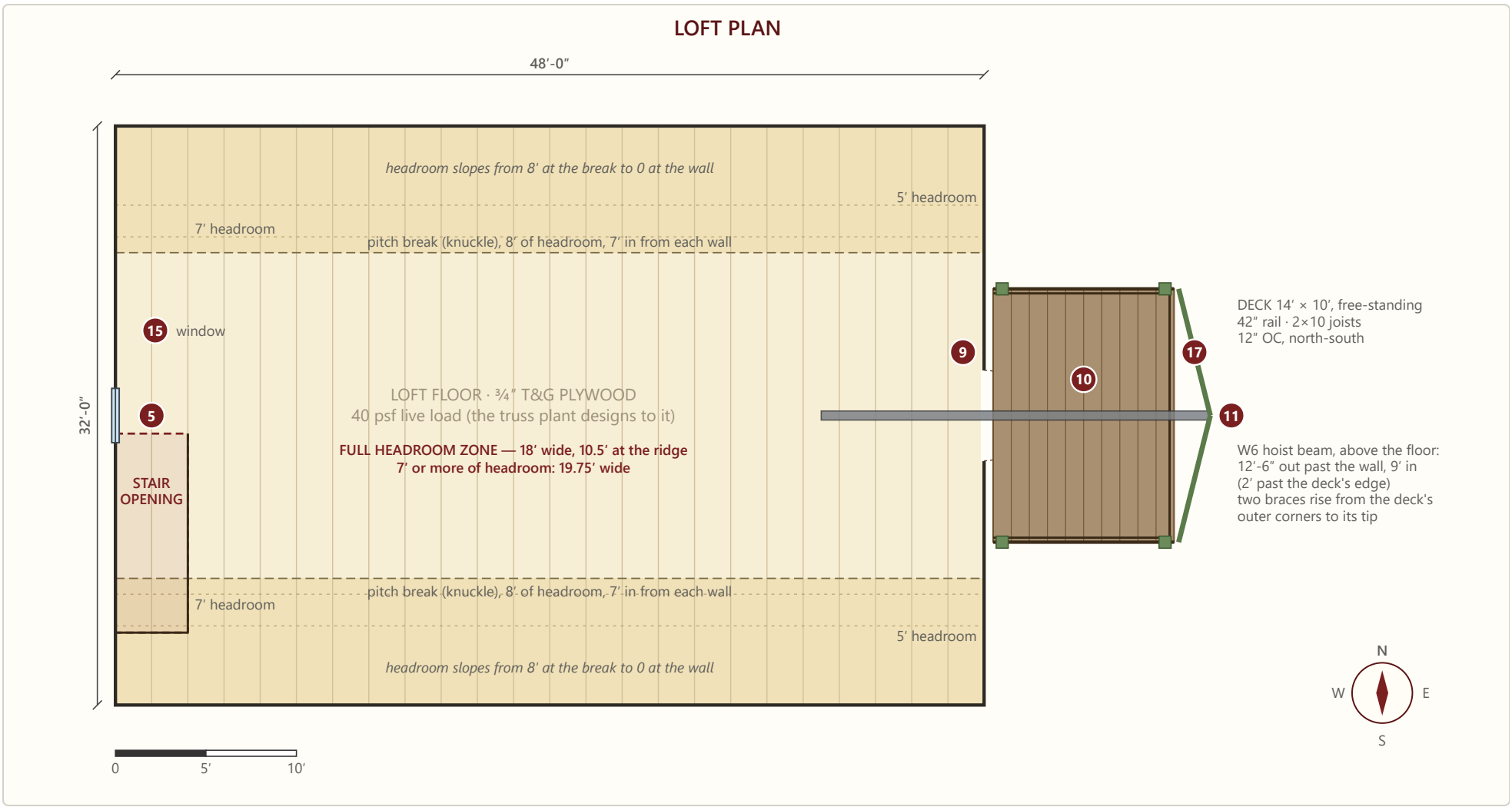
The same frame from the northwest: the north wall with its three windows (marker 15), and the west gable with its three posts, the loft window, and the studs of the gable truss, which stop at its collar tie. The trusses' tails and the purlins run past the gables by the 1 ft rake overhang, and the deck and hoist show at the far end.

Floor Plan (Main Level)



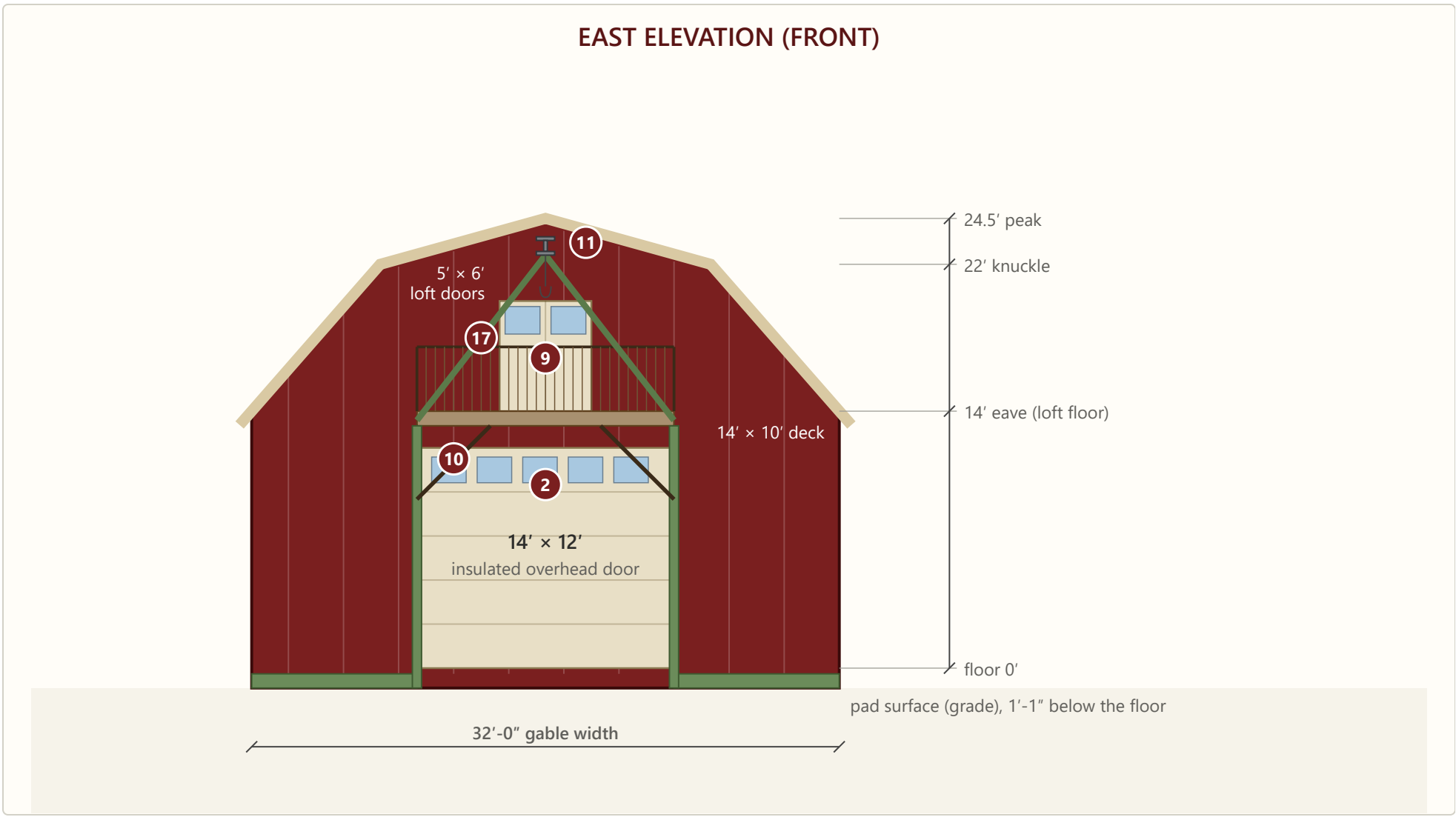
Floor plan, main level. Posts are filled squares, drawn larger than they are. The windows (marker 15) sit in the middle of their bays: three in the north wall and three in the south wall. The dashed outline over the apron is the deck above. The stair's second flight climbs north along the west wall to the dashed opening in the loft floor. The east wall dimensions are the 9′ + 14′ door + 9′ that add up to the 32′ width.

Loft Plan



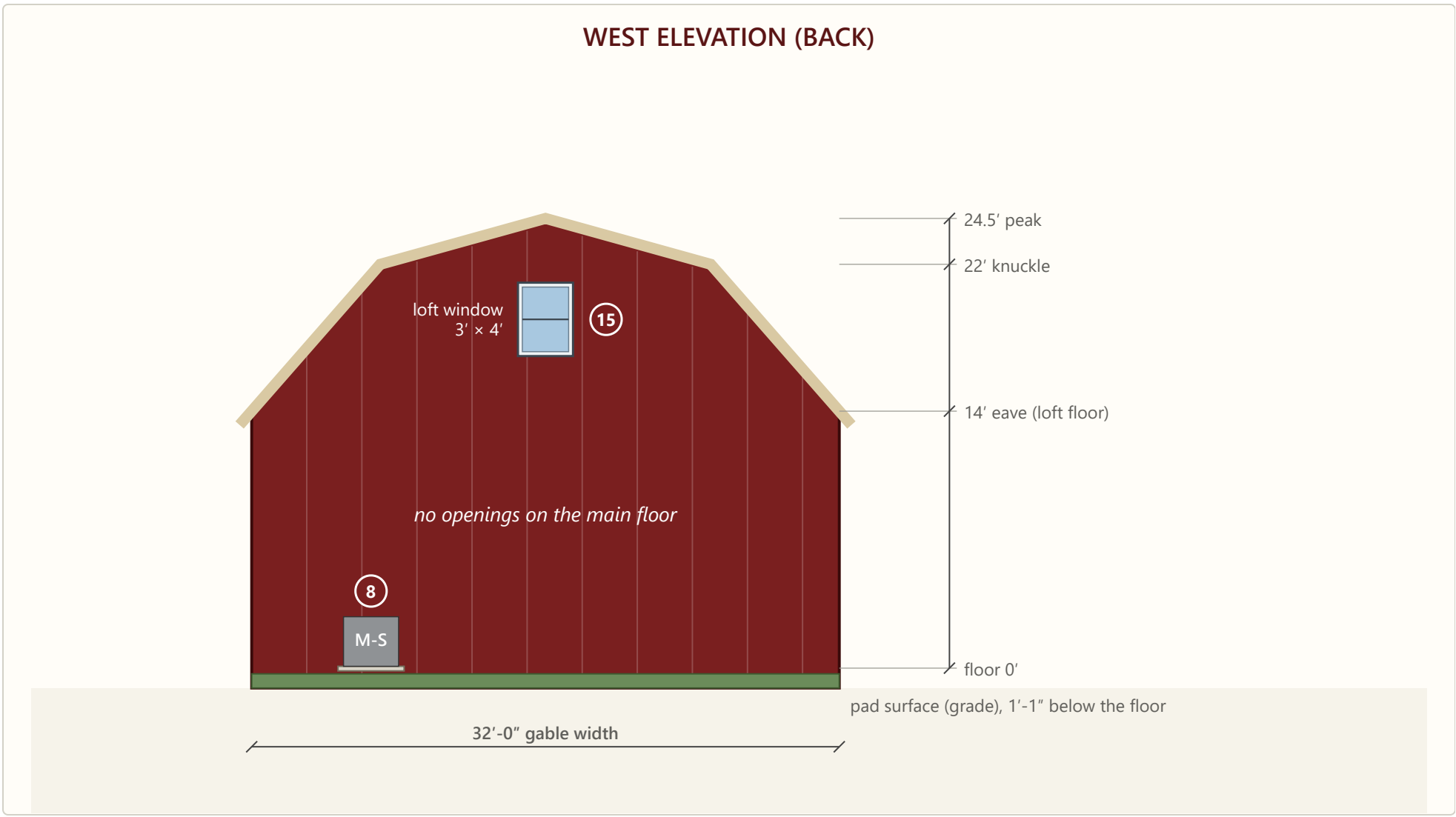
Loft plan. The stair opening is at the west wall, with the loft window centered in the west wall beside it; the loft double doors and the 14′ × 10′ landing deck are at the east end, and the hoist beam reaches 12′-6″ past the wall, 2 ft beyond the deck's outer edge, with two braces (marker 17) from the deck's outer corners to its tip. Headroom falls from the pitch-break line toward each sidewall: about 950 sq ft of the loft has 7 ft or more of headroom, measured to the roof line (the top of the truss chords). Under 7¼″ top chords the clear height is 11″ lower at the pitch break and 7½″ lower at the ridge (see the truss package).

East Elevation (Front)



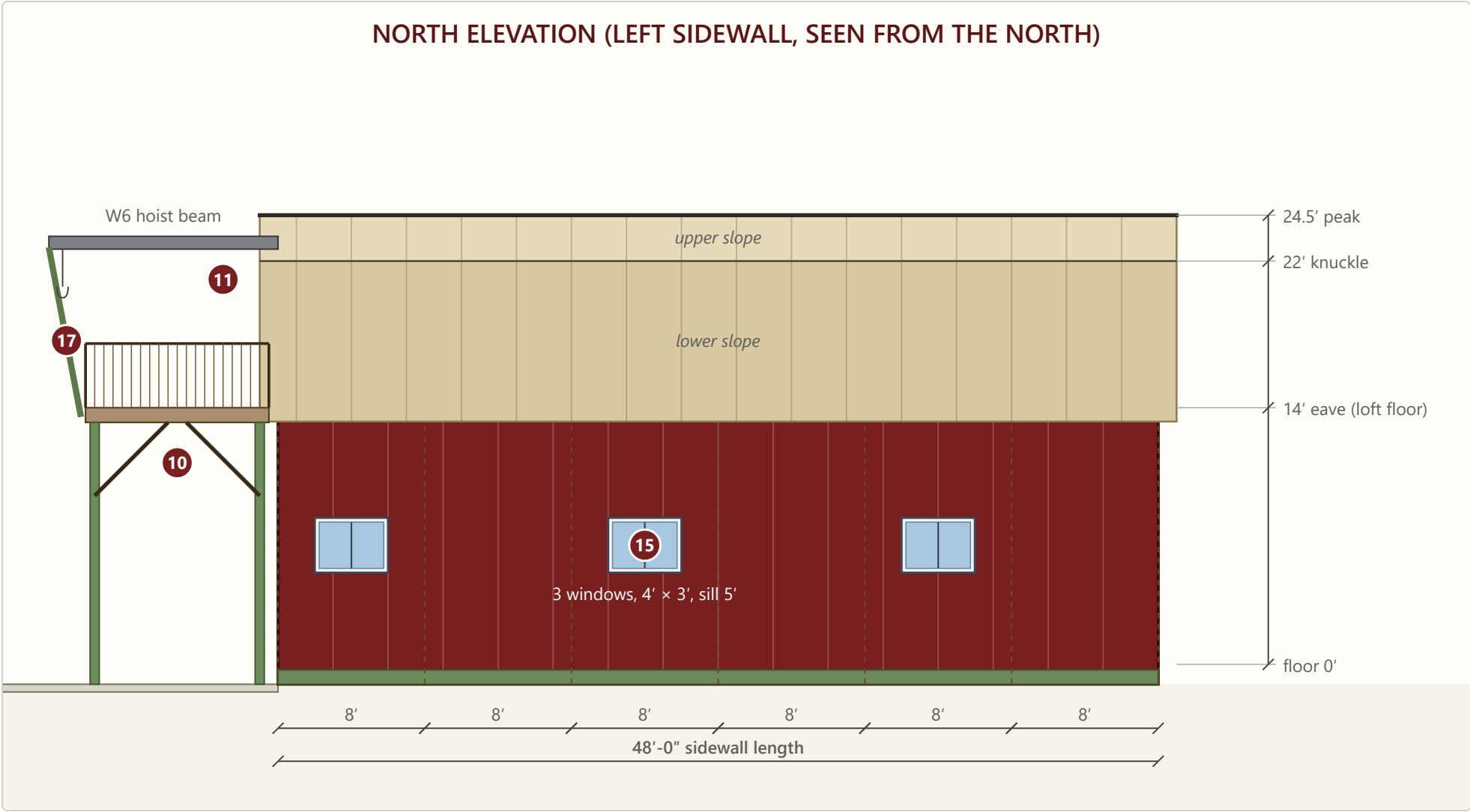
East (front) elevation. The gambrel gable wall with the 14′ × 12′ overhead door, the loft double doors above it, the 14′ × 10′ deck on free-standing posts, and the hoist beam seen end-on, with its two braces (marker 17) rising in a V from the deck's outer corners to its tip and the chain hanging between them. The deck posts stand on the door-jamb lines.

West Elevation (Back)



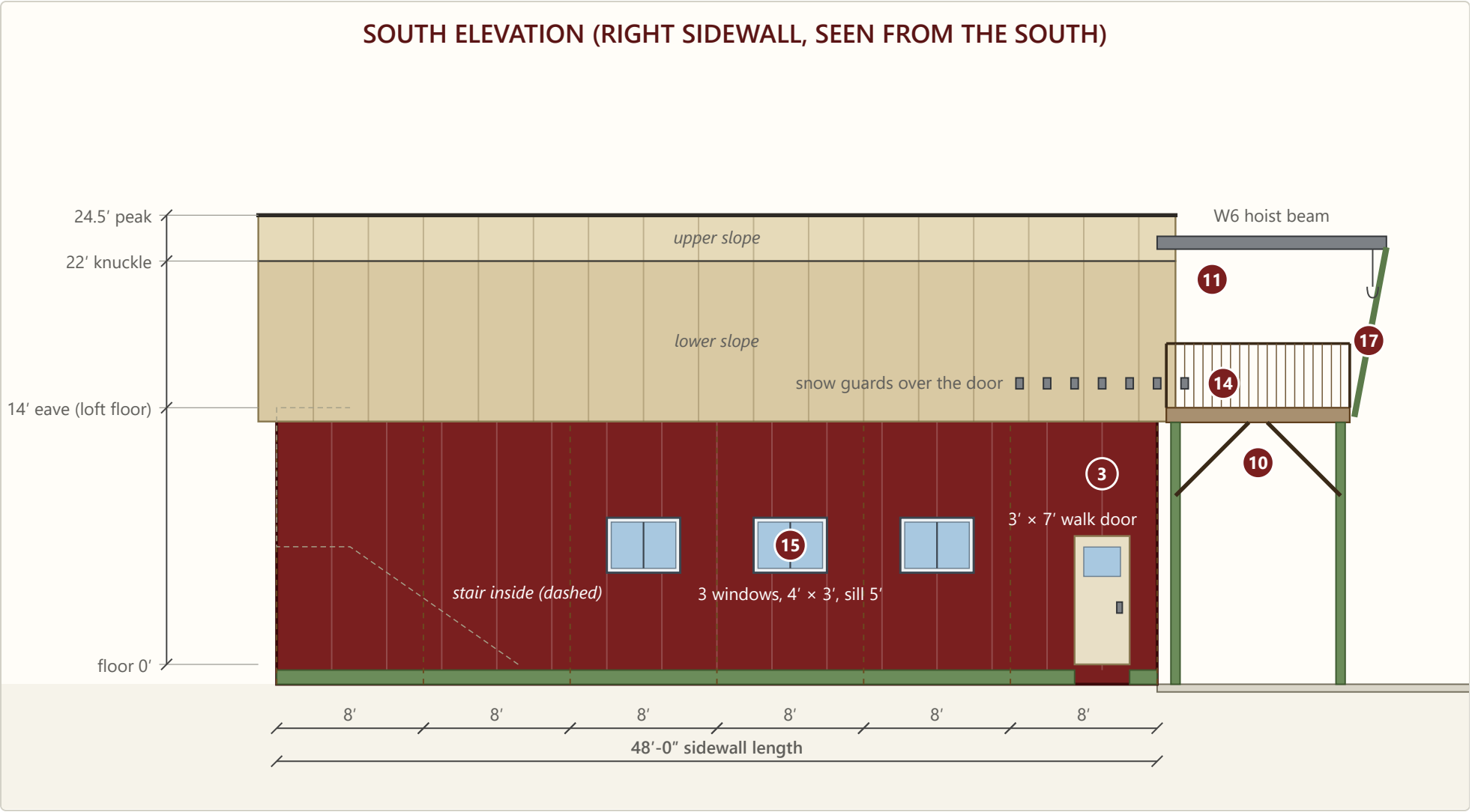
West (back) elevation. A gambrel gable with one window at the loft (marker 15), centered, and no openings on the main floor. The mini-split condenser is shown at the base; the owner may put it on the south wall instead.

North Elevation (Left Sidewall)

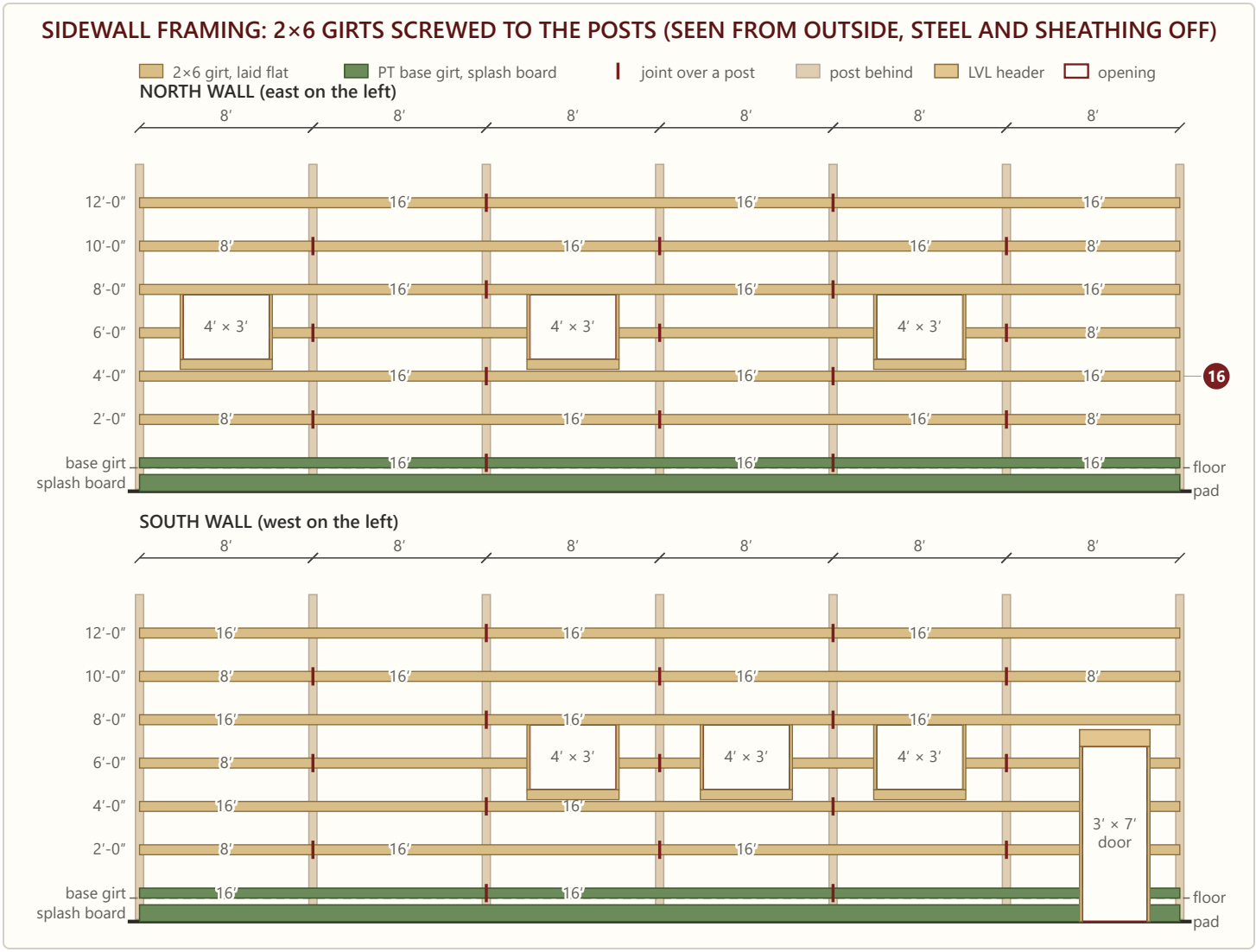


North sidewall, seen from the north, so east is on the left. Three windows (marker 15), in bays 2, 4, and 6 counted from the west end; the rest is plain steel. The dashed lines are the post locations behind the siding. The deck, the hoist beam, and one of its braces show at the east end.

South Elevation (Right Sidewall)



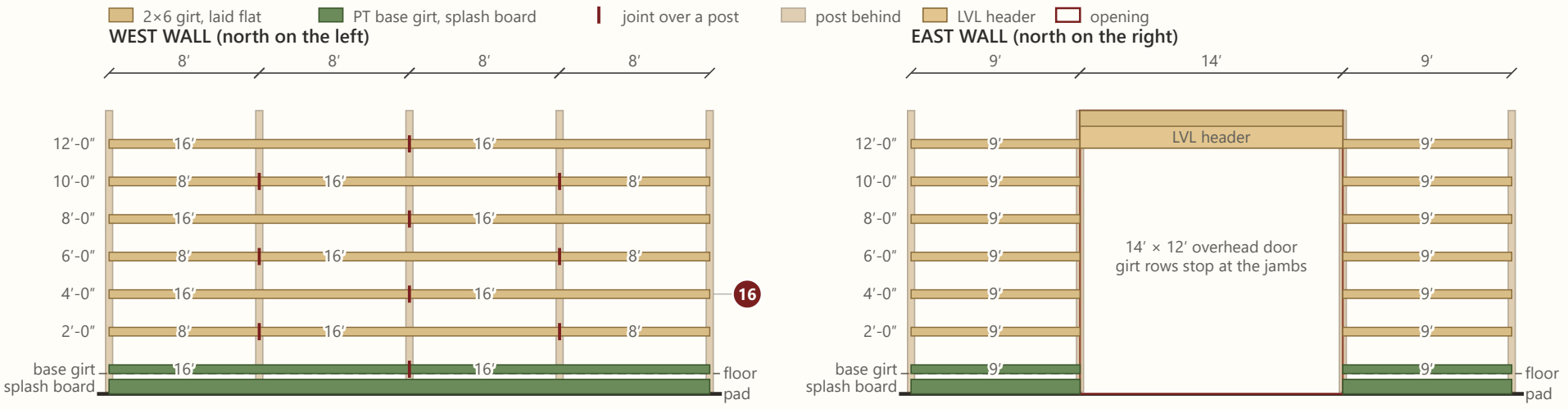
South sidewall, seen from the south, so west is on the left. Three windows (marker 15) in bays 3, 4, and 5, with none at the west end, where the stair is. The 3070 walk door is near the east end, well clear of the first stair flight (dashed, inside) and of the last window. The snow guards sit on the lower slope over the door.



The 2×6 girts (marker 16) on the north and south walls, seen from outside with the steel, furring, and sheathing taken off; the posts are behind them. Each board is laid flat on the posts and screwed to every post it crosses. The posts are 8 ft apart, so a 16 ft board spans two bays and every joint falls on a post. Rows alternate between two patterns, 16-16-16 and 8-16-16-8 (each pair of 8 ft pieces is one 16 ft board cut in half), so neighboring rows never share a joint post. At each window the 6′ row is cut: the girt at 8′ is the header, and a sill and two jack studs frame the opening. At the walk door the base girt and the 2′, 4′, and 6′ rows stop at jack studs beside the door. Row labels give the bottom edge of each row above the floor. The base girt is pressure-treated and sits at the floor line, with the 2×10 PT splash board screwed to the posts below it.

Wall Framing: East and West Walls

END WALL FRAMING: 2×6 GIRTS SCREWED TO THE POSTS (SEEN FROM OUTSIDE, STEEL AND SHEATHING OFF)



The same girts on the end walls, seen from outside. The west wall uses the same two patterns across its 32 ft: 16-16 and 8-16-8. On the east wall the door jambs leave a 9 ft run on each side of the door in every row, and each pair of 9 ft pieces is one 18 ft board cut in half. The LVL header carries the loft floor over the door, with the gable truss's bottom chord above it.

Mountain Prairie Acres · Workshop pole barn plans and elevations · updated October 23, 2026 · drawn from `docs/workshop_model.js`.